IX. Mini Project – Student Performance Analysis

AIM:

To clean, process, analyze, and visualize student academic data using Python, Power BI, and Tableau to identify performance trends, subject-wise results, attendance patterns, and the relationship between study hours and scores.

DOMAIN:

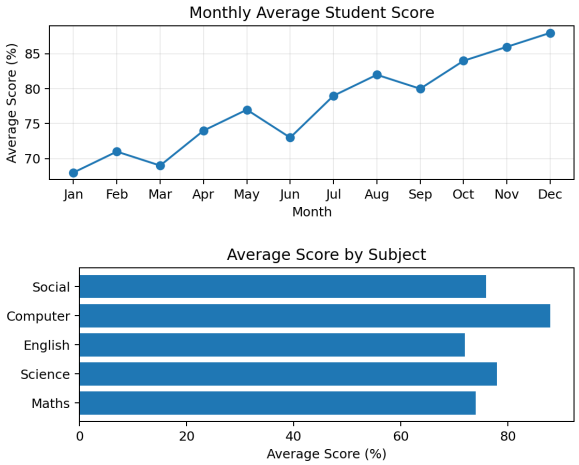
Educational Data Analytics

PROBLEM STATEMENT:

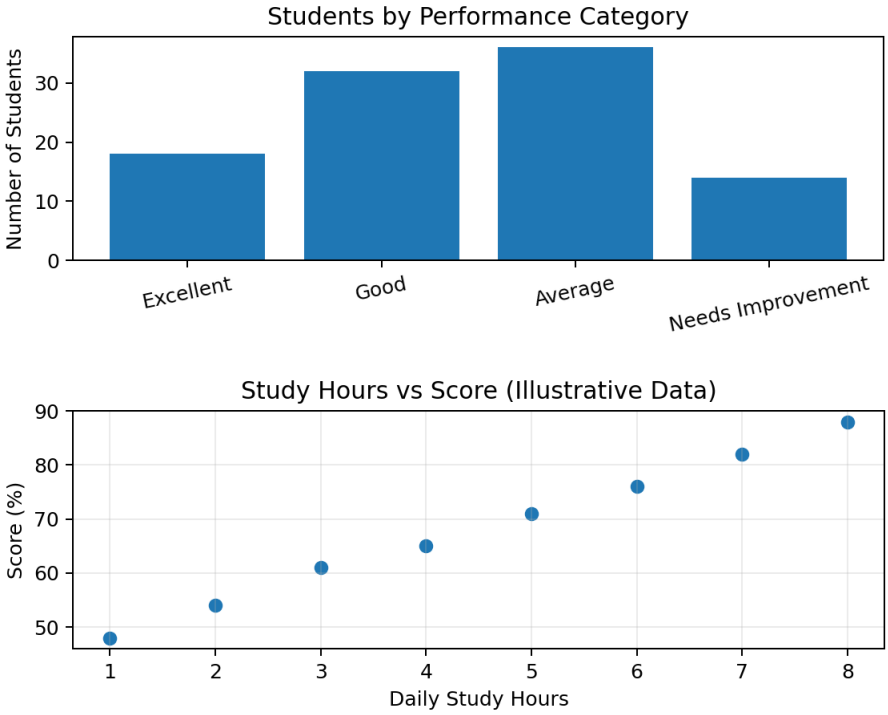
Educational institutions collect marks, attendance, and study-related data, but it can be difficult to identify students who need support and understand overall academic trends. A data-driven dashboard can compare performance across subjects and months, summarize performance categories, and help educators make informed decisions.

OUTPUT:

Python:



Monthly average scores and subject-wise score visualizations generated using Python. Values shown are illustrative sample data.

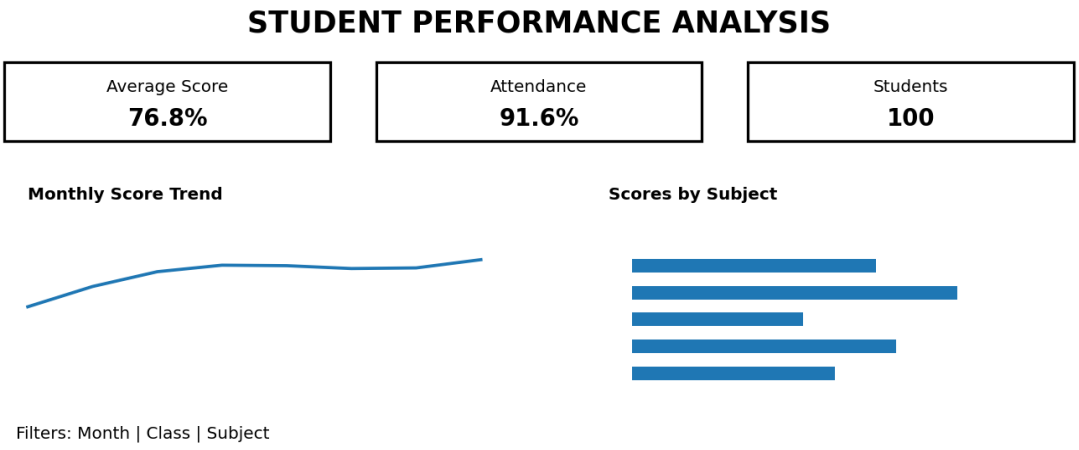


Performance-category and study-hours comparison charts generated using Python. Values shown are illustrative sample data.

Power BI:

Create an interactive dashboard with KPI cards for average score, attendance, and number of students; slicers for subject, class, and month; and charts for subject-wise scores, monthly trends, and performance categories.

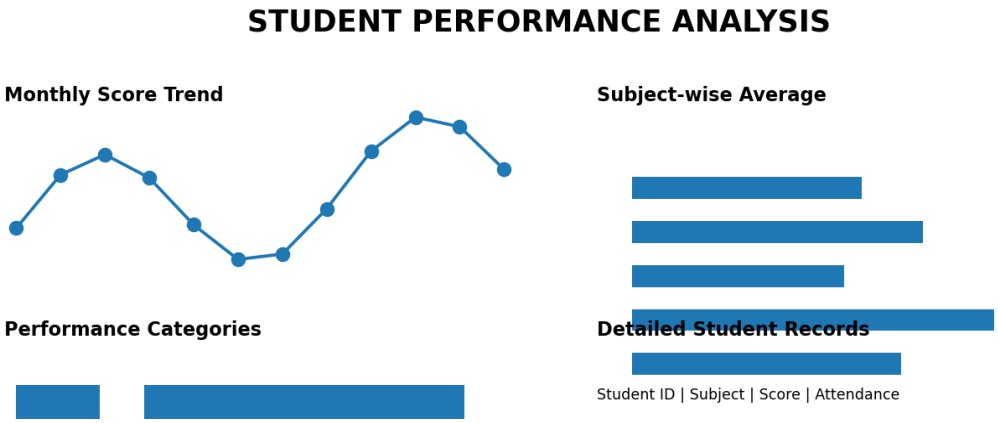
Suggested Power BI dashboard layout:



Illustrative Power BI dashboard design for Student Performance Analysis.

Tableau:

In Tableau, build a dashboard with a monthly performance line chart, subject-wise bar chart, attendance summary, and a detailed student table. Add filters for class, subject, and month to explore patterns interactively.



Illustrative Tableau dashboard design for Student Performance Analysis.

RESULT:

Thus, student academic data was organized, analyzed, and visualized using Python, Power BI, and Tableau. The charts and proposed dashboards show how monthly trends, subject-wise performance, attendance, and study-hour patterns can be explored. The sample values are illustrative and should be replaced with actual institutional data for real analysis.